

Problem

Synthesize the function

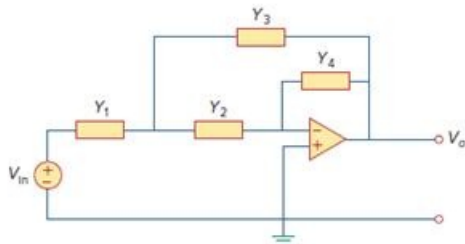
$$\frac{V_o(s)}{V_{in}} = \frac{-2s}{s^2 + 6s + 10}$$

using the op amp circuit shown in Fig. 16.34. Select

$$Y_1 = \frac{1}{R_1}, \quad Y_2 = sC_1, \quad Y_3 = sC_2, \quad Y_4 = \frac{1}{R_2}$$

Let $R_1 = 1 \text{ k}\Omega$, and determine C_1 , C_2 , and R_2 .

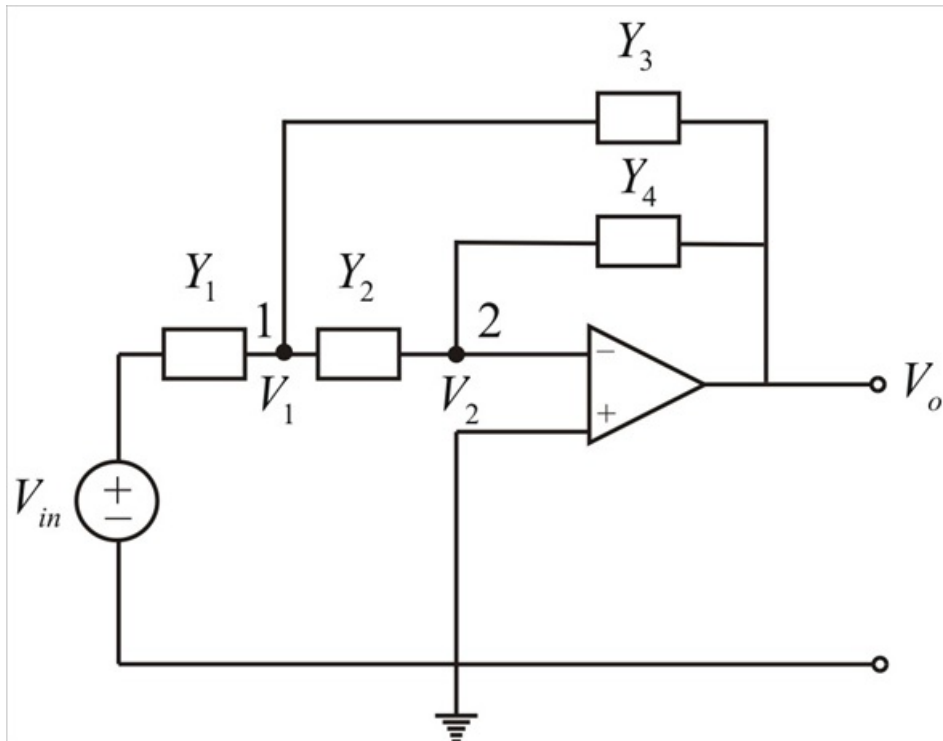
Figure 16.34:



Step-by-step solution

Step 1 of 5

Given circuit



Step 2 of 5

Using Nodal analysis, we get

At node 1

$$(V_1 - V_{in})Y_1 + (V_1 - V_o)Y_3 + (V_1 - V_2)Y_2 = 0$$

As we know that

$$V^- = V^+ = 0$$

Where $V^- = V_2 \Rightarrow 0$

$$(V_1 - V_{in})Y_1 + (V_1 - V_o)Y_3 + V_1Y_2 = 0$$

$$V_1(Y_1 + Y_3 + Y_2) = V_{in}Y_1 + V_oY_3 \text{ ---- (1)}$$

At node 2

$$(V_2 - V_1)Y_2 + (V_2 - V_o)Y_4 = 0$$

$$-V_1Y_2 = V_oY_4$$

$$V_1 = \frac{-V_oY_4}{Y_2}$$

Step 3 of 5

Substituting V_1 value in equation (1), we get

$$\left(\frac{-V_oY_4}{Y_2} \right) (Y_1 + Y_3 + Y_2) = V_{in}Y_1 + V_oY_3$$

$$-V_o(Y_1Y_4 + Y_3Y_4 + Y_2Y_4) - Y_3Y_2V_o = V_{in}Y_1Y_2$$

$$-V_o(Y_1Y_4 + Y_2Y_4 + Y_3Y_4 + Y_2Y_3) = V_{in}Y_1Y_2$$

Therefore, the function

$$\frac{V_o(s)}{V_{in}(s)} = \frac{-Y_1Y_2}{Y_1Y_4 + Y_2Y_4 + Y_3Y_4 + Y_2Y_3}$$

Step 4 of 5

We select

$$Y_1 = \frac{1}{R_1}, \quad Y_2 = sC$$

$$Y_3 = sC_2, \quad Y_4 = \frac{1}{R_2}$$

$$\frac{V_o(s)}{V_{in}(s)} = \frac{-\left(\frac{1}{R_1}\right)(sC_1)}{\left(\frac{1}{R_1}\right)\left(\frac{1}{R_2}\right) + (sC_1)(sC_2) + (sC_1)\left(\frac{1}{R_2}\right) + (sC_2)\left(\frac{1}{R_2}\right)}$$

$$\frac{V_o(s)}{V_{in}(s)} = \frac{-\frac{sC_1}{R_1}}{s^2C_1C_2 + \frac{s}{R_2}(C_1 + C_2) + \frac{1}{R_1R_2}}$$

$$\frac{V_o(s)}{V_{in}(s)} = \frac{-\frac{s}{R_1C_2}}{s^2 + \frac{s}{R_2}\left(\frac{1}{C_2} + \frac{1}{C_1}\right) + \frac{1}{R_1R_2C_1C_2}}$$

Step 5 of 5

Compare by the given function and substitute the value of R_1 as $1\text{ k}\Omega$

Thus, the given function is

$$\frac{V_o(s)}{V_{in}(s)} = \frac{-2s}{s^2 + 6s + 10}$$

$$\frac{1}{R_1 C_2} = 2$$

$$\frac{1}{(1\text{k})C_2} = 2$$

$$C_2 = \frac{1}{2\text{k}}$$

$$\therefore \boxed{C_2 = 0.5\text{ mF}}$$

$$\frac{1}{R_2} \left(\frac{1}{C_2} + \frac{1}{C_1} \right) = 6$$

$$\frac{C_1 + C_2}{C_1 C_2 R_2} = 6$$

$$C_1 + 0.5\text{ m} = 6(0.5\text{ m}) R_2 C_1$$

$$C_1 (1 - 3\text{ m} R_2) = -0.5\text{ m}$$

$$\therefore C_1 = \frac{-0.5\text{ m}}{1 - 3\text{ m} R_2}$$

$$\frac{1}{R_1 R_2 C_1 C_2} = 10$$

$$(1\text{k})(0.5\text{ m}) R_2 C_1 = 0.1$$

$$R_2 C_1 = 0.2$$

$$R_2 \left(\frac{-0.5\text{ m}}{1 - 3\text{ m} R_2} \right) = 0.2$$

$$-0.5\text{ m} R_2 = 0.2 - 0.6\text{ m} R_2$$

$$0.1\text{ m} R_2 = 0.2$$

$$\therefore \boxed{R_2 = 2\text{ k}\Omega}$$

Then

$$C_1 = \frac{-0.5\text{ m}}{1 - (3\text{ m})(2\text{k})}$$

$$C_1 = \frac{-0.5\text{ m}}{-5}$$

$$\therefore \boxed{C_1 = 0.1\text{ mF}}$$